

MethodsX

BDNeuro-MRI: A Bangladeshi Clinical Brain Tumor MRI Dataset for Four-Class Deep Learning Classification

--Manuscript Draft--

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Abstract:	This dataset contains 5,941 brain MRI images spanning four diagnostic classes: glioma, meningioma, pituitary tumor, and no tumor, collected from Epic Health Care, Chittagong, Bangladesh (Ethics Ref: EHC/CARD/2025/0047) under the supervision of a consultant neurologist. From an initial pool of 11,148 clinically sourced images, 833 exact duplicates (MD5 hash) and 4,374 near-duplicates (perceptual hashing, Hamming distance ≤ 5) were identified and removed before splitting, eliminating hidden train/test leakage largely unaddressed in existing public brain tumor MRI collections. All images were fully anonymized prior to release. The cleaned dataset is partitioned into training (4,160 images, 70%), validation (892 images, 15%), and test (889 images, 15%) sets using per-class stratified sampling, with zero cross-split overlap confirmed by content hash. A documented preprocessing pipeline resizes all images to 224×224 pixels, applies training-only augmentation (rotation, flipping, affine translation, and resized cropping), and performs channel-wise normalization. Seven deep-learning architectures, including CNN, Vision Transformer, Hybrid CNN–ViT, and frozen/fine-tuned ResNet50 and DenseNet121, were benchmarked to confirm the dataset is learnable and leakage-free. The dataset follows the ImageFolder convention and is publicly available on Mendeley Data (DOI: 10.17632/zwr4ntf94j.7). To our knowledge, this is the first duplicate-verified, institutionally sourced brain tumor MRI dataset from a Bangladeshi clinical setting.
Opposed Reviewers:	
Response to Reviewers:	We thank the Editor and Reviewer #1 for their constructive comments, which centered on making the novelty and contribution of the work more explicit. We have revised the manuscript accordingly. Please find our point-by-point responses below; all corresponding changes are marked using Track Changes in the revised manuscript file. Reviewer #1, Comment 1: The paper should clearly highlight the novelty and unique contributions of the proposed work. Response: We have added an explicit novelty statement clarifying that the contribution

of this paper is not a new deep-learning architecture, but a fully documented, auditable curation protocol for clinical MRI datasets combining MD5 exact-duplicate screening, perceptual near-duplicate screening, and post-split content-hash leakage verification, applied for the first time to a Bangladeshi clinical brain tumor MRI pool.
Change: New paragraph added at the end of the Background section (tracked insertion).

Reviewer #1, Comment 2: Some public MRI datasets are already screened for duplicates using MD5 hashing; please clarify how this work differs and what it adds.
Response: We clarified that exact-hash (MD5) screening alone cannot detect re-exported, re-compressed, or cropped near-duplicate images, and that, to our knowledge, no existing MD5-screened public collection additionally verifies zero cross-split overlap via a post-split content-hash check, which is the specific gap this method closes.
Change: New sentence added in the Background section (tracked insertion).

Reviewer #1, Comment 3: After the hashing step, the pipeline follows a standard deep learning workflow; please clarify the methodological novelty beyond hashing.
Response: We clarified that the seven benchmarking architectures are deliberately trained using standard, unmodified pipelines, serving only as a diagnostic instrument to confirm the curated splits are learnable and leakage-free. We state explicitly that no architectural novelty is claimed at this stage, and that the paper's novelty lies entirely in the upstream duplicate-screening and leakage-verification protocol.
Change: New paragraph added in the "Downstream benchmarking" subsection (tracked insertion).

We believe these revisions fully address the reviewer's concerns and thank the Editor and Reviewer for helping us sharpen this distinction.

Detailed Response to Reviewers

Manuscript MEX-D-26-01809

Manuscript title: BDNeuro-MRI: A Bangladeshi Clinical Brain Tumor MRI Dataset for Four-Class Deep Learning Classification

Journal: MethodsX

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We thank the Editor and Reviewer #1 for their careful reading of the manuscript and for the constructive comments, all of which centered on making the novelty and contribution of the work more explicit. We have revised the manuscript accordingly. Point-by-point responses are given below; all changes are marked using Word's Track Changes function in the accompanying revised manuscript file.

Reviewer #1

Comment 1

Reviewer comment: The paper should clearly highlight the novelty and unique contributions of the proposed work.

Response: We agree and have added an explicit novelty/contribution statement. We clarify that the contribution of this paper is not a new deep-learning architecture, but a fully documented, auditable curation protocol for clinical MRI datasets that combines (i) MD5 exact-duplicate screening, (ii) perceptual-hash near-duplicate screening (Hamming distance ≤ 5), and (iii) an explicit post-split content-hash verification step confirming zero train/validation/test overlap applied for the first time to a Bangladeshi clinical brain tumor MRI pool. We also clarify that every screening decision is auditable via retained intermediate hash tables, and that the released BDNeuro-MRI dataset is offered as a concrete, benchmarked instance of this protocol.

Change made: A new paragraph has been added at the end of the Background section, immediately before the “Method details” heading, stating this novelty explicitly (tracked insertion, revised manuscript, Background section).

Comment 2

Reviewer comment: A few publicly available MRI datasets have already been screened for duplicate images using MD5 hashing. Please clearly describe how the proposed work differs from these existing approaches and what additional contribution it provides.

Response: We agree that this distinction was not stated clearly enough in the original submission. We have added text acknowledging that some existing public MRI collections do report MD5-based exact-duplicate removal, but noting that exact-hash matching alone cannot detect re-exported, re-compressed, or minimally cropped copies of the same underlying scan precisely the near-duplicate cases most likely to cause train/test leakage. We further clarify that, to our knowledge, none of these MD5-screened collections additionally verify zero cross-split overlap via a dedicated post-split content-hash check, which is the specific gap our method is designed to close.

Change made: A new sentence has been inserted in the Background section, immediately after the sentence describing limitations of existing public brain tumor MRI collections (“...which complicates reproducibility and fair comparison across studies.”) and before “This motivated us to develop...” (tracked insertion, revised manuscript, Background section).

Comment 3

Reviewer comment: After the hashing step, the methodology follows a standard deep learning workflow. The paper should clarify the methodological novelty beyond the hashing process and explain how the proposed approach differs from conventional deep learning pipelines.

Response: We confirm that the seven benchmarking architectures (custom CNN, ViT, Hybrid CNN–ViT, and frozen/fine-tuned ResNet50 and DenseNet121) are deliberately trained using standard, unmodified deep-learning pipelines, with no architecture-specific tuning. We have clarified in the manuscript that this is an intentional design choice rather than an oversight: the conventional pipeline serves as a fixed external instrument to test whether the duplicate-screened, leakage-verified splits are learnable and free of leakage-inflated performance. We explicitly state that no claim of architectural novelty is made for the benchmarking stage, and that the methodological novelty of the paper lies entirely upstream, in the duplicate/near-duplicate screening and post-split leakage-verification protocol.

Change made: A new paragraph has been added in the “Downstream benchmarking” subsection (Method validation), immediately after the sentence listing the seven architectures and before Table 4 (tracked insertion, revised manuscript, Downstream benchmarking section).

We believe these revisions directly address all three points raised by Reviewer #1 and clarify that the contribution of this work lies in the documented, auditable dataset curation and leakage-verification protocol, rather than in the downstream deep-learning architectures, which are intentionally kept conventional to isolate the effect of data quality. We thank the Editor and Reviewer for helping us sharpen this distinction, and hope the revised manuscript is now suitable for publication in MethodsX.



Article information

Article title

BDNeuro-MRI: A Bangladeshi Clinical Brain Tumor MRI Dataset for Four-Class Deep Learning Classification

Authors

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Keywords

Brain tumor classification; MRI; dataset curation; duplicate screening; convolutional neural networks; vision transformers; transfer learning; deep learning; medical imaging

Related research article

None.

Abstract

This dataset contains 5,941 brain MRI images spanning four diagnostic classes: glioma, meningioma, pituitary tumor, and no tumor, collected from Epic Health Care, Chittagong, Bangladesh (Ethics Ref: EHC/CARD/2025/0047) under the supervision of a consultant neurologist. From an initial pool of 11,148 clinically sourced images, 833 exact duplicates (MD5 hash) and 4,374 near-duplicates (perceptual hashing, Hamming distance ≤ 5) were identified and removed before splitting, eliminating hidden train/test leakage largely unaddressed in existing public brain tumor MRI collections. All images were fully anonymized prior to release. The cleaned dataset is partitioned into training (4,160 images, 70%), validation (892 images, 15%), and test (889 images, 15%) sets using per-class stratified sampling, with zero cross-split overlap confirmed by content hash. A documented preprocessing pipeline resizes all images to 224×224 pixels, applies training-only augmentation (rotation, flipping, affine translation, and resized cropping), and performs channel-wise normalization. Seven deep-learning architectures, including CNN, Vision Transformer, Hybrid CNN-ViT, and frozen/fine-tuned ResNet50 and DenseNet121, were benchmarked to confirm the dataset is learnable and leakage-free. The dataset follows the ImageFolder convention and is publicly available on Mendeley Data (DOI: 10.17632/zwr4ntf94j.7). To our knowledge, this is the first duplicate-verified, institutionally sourced brain tumor MRI dataset from a Bangladeshi clinical setting.

Graphical abstract

Submitted as a separate file (see the “Graphical Abstract” item type in the manuscript submission system). It depicts the six-stage pipeline: clinical MRI collection → duplicate screening (MD5 + perceptual hash) → stratified 70/15/15

split with overlap verification → preprocessing (resize/tensor/normalize) → training-only augmentation → benchmarking of seven deep-learning architectures.

Specifications table

Subject area	Computer Science
More specific subject area	Medical image data curation; brain tumor MRI classification; deep learning; duplicate/leakage detection in imaging datasets
Name of your method	Duplicate-Screened Clinical MRI Curation and Preprocessing Pipeline (DS-CMCP)
Name and reference of original method	Not applicable — the pipeline customizes standard duplicate-detection (MD5 hashing; perceptual/difference hashing) and torchvision-based preprocessing techniques for clinical brain tumor MRI curation; no single prior pipeline is being reproduced.
Resource availability	Curated dataset (raw images, splits, and pretrained model weights): Mendeley Data, DOI 10.17632/zwr4ntf94j.7 , https://data.mendeley.com/datasets/zwr4ntf94j/7 (no registration required). Preprocessing implemented in Python 3 using torchvision.transforms and PyTorch DataLoader.

Background

Automated deep-learning classification of brain MRI into diagnostic categories such as glioma, meningioma, pituitary tumor, and no tumor is increasingly used as a decision-support tool, with recent transfer-learning and attention-based pipelines reporting accuracies above 96–98% on four-class tasks [1–3]. However, the reliability of these results depends critically on how the underlying dataset was curated. A particularly under-reported failure mode is hidden train/test leakage caused by exact or near-duplicate images re-exported, re-compressed, or minimally cropped copies of the same scan being distributed across both splits of a released dataset, which can inflate reported performance without reflecting genuine generalization. Systematic reviews of the field converge on related, unresolved issues: dataset bias, limited cross-institutional generalizability, and inconsistent evaluation protocols [4–6]. Even newly released, expertly annotated resources illustrate the broader gap: the recently introduced BRISC dataset adds radiologist-verified segmentation masks, but its scans were collated from multiple pre-existing public repositories that originally lacked those labels, rather than collected prospectively from a defined clinical population [7]. More broadly, many public brain tumor MRI collections are distributed with heterogeneous resolutions, inconsistent folder structures, undocumented duplicate content, and minimal documentation of curation and splitting procedures [8–11], which complicates reproducibility and fair comparison across studies. A small number of existing public MRI collections do report an MD5-based exact-duplicate removal step; however, exact-hash matching alone does not catch re-exported, re-compressed, or minimally cropped copies of the same underlying scan — precisely the near-duplicate cases most likely to leak across splits — and, to our knowledge, none of these MD5-screened collections additionally verify zero cross-split overlap via a dedicated post-split content-hash check. This gap between being exact-duplicate screened and being leakage-verified is what the present method is designed to close. This motivated us to develop and document a concrete, step-by-step curation method covering clinical sourcing under ethics clearance, systematic duplicate screening, stratified splitting with leakage verification, and a fully specified preprocessing/augmentation pipeline that other groups can apply when building similar clinical imaging datasets, whether or not they use our

specific dataset. We demonstrate the method on a Bangladeshi clinical MRI pool, producing the BDNeuro-MRI dataset as a worked example and validating it with seven benchmarked deep-learning architectures.

To state the novelty of this work explicitly: the contribution is not a new deep-learning architecture, but a fully documented, auditable curation protocol for clinical MRI datasets — one that combines (i) MD5 exact-duplicate screening, (ii) perceptual-hash near-duplicate screening (Hamming distance ≤ 5), and (iii) an explicit post-split content-hash verification step confirming zero train/validation/test overlap, applied for the first time to a Bangladeshi clinical brain tumor MRI pool. Unlike prior public brain tumor MRI collections, which are typically released with undocumented or absent duplicate-screening procedures, every screening decision in this pipeline is auditable via retained intermediate hash tables (Method details, Step 2), and the resulting BDNeuro-MRI dataset is offered as a concrete, benchmarked instance of that protocol that other groups can reuse or adapt.

Method details

Our goal is to publish methods that can be replicated by other researchers; this section gives the full step-by-step procedure.

1. Clinical data sourcing

MRI images were prospectively collected from Epic Health Care, 19 K.B. Fazlul Kader Road, Panchlish, Chittagong 4000, Bangladesh, under institutional ethics clearance (Ref: EHC/CARD/2025/0047), with the direct supervision of a consultant neurologist (MBBS, BCS (Health), MD (Neurology)), Department of Neuromedicine, over the period September 2025 – April 2026. Images were retained in their original JPEG or PNG format and organized by diagnostic class label: glioma, meningioma, pituitary tumor, or no tumor (healthy/control). Prior to release, every image was screened for completeness and fully anonymized, with no personally identifiable information retained, in accordance with institutional data-protection and patient-confidentiality obligations. This stage produced an initial pool of 11,148 clinically sourced images.

2. Duplicate and near-duplicate screening

Because clinical image pools are frequently re-exported or re-compressed at multiple points in the acquisition/storage workflow, the same underlying scan can appear more than once. Left unaddressed, this causes hidden train/test leakage once the pool is split. The method screens the full pool in two passes before any splitting occurs:

- **Exact-duplicate screening:** an MD5 content hash is computed for every image; byte-identical files are flagged and all but one copy are removed. This step identified and removed 833 exact duplicates.
- **Near-duplicate screening:** a perceptual (difference) hash is computed for every remaining image; pairs with Hamming distance ≤ 5 are treated as near-duplicates and redundant copies are removed. This step identified and removed 4,374 near-duplicates.

Together, these two passes reduced the initial pool of 11,148 images to 5,941 verified-unique images (5,207 images removed in total). Researchers applying this method to their own clinical pools should retain the intermediate hash tables, as they allow the screening decision for any given image to be audited after the fact.

3. Stratified splitting with cross-split leakage verification

The 5,941 verified-unique images were divided into training, validation, and test sets using per-class stratified sampling targeting a 70%/15%/15% allocation, so that class proportions are preserved within each split (Table 1). After splitting, a content-hash comparison was run across every pair of splits to confirm zero image overlap. This explicit verification step is recommended as a mandatory check whenever duplicate screening has been performed upstream.

Class	Train	Val	Test	Total
Glioma	1534	329	328	2191
Meningioma	1035	222	221	1478
No Tumor	551	118	118	787
Pituitary	1040	223	222	1485
Total	4160	892	889	5941

Table 1. Class-wise distribution of images across the training, validation, and test splits produced by Step 3, after the duplicate screening in Step 2. The dataset contains 5,941 images in total, partitioned approximately 70%:15%:15%, with class proportions preserved across all three splits and zero cross-split overlap confirmed by content hash.

4. Preprocessing pipeline

All retained images are passed through a standardized four-stage preprocessing pipeline, implemented in Python using the torchvision.transforms module and applied on-the-fly via a PyTorch DataLoader (batch size 16, four parallel worker processes). The pipeline is illustrated in Figure 1.

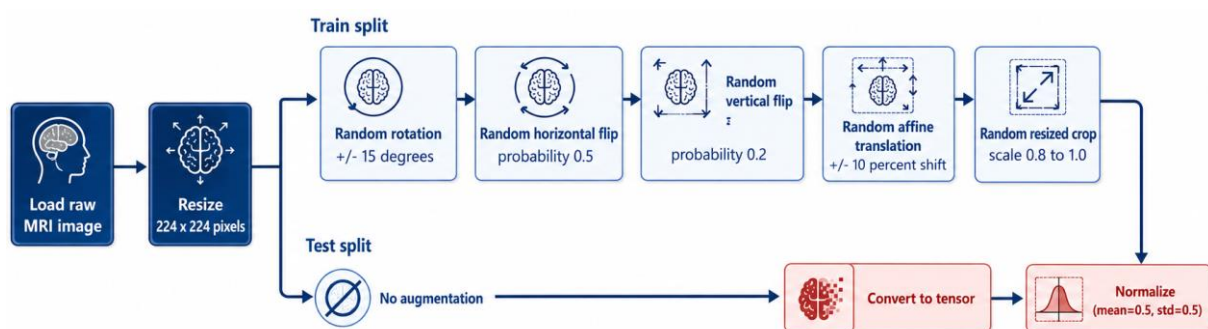


Figure 1. The four-stage preprocessing pipeline: resize → augment (training split only) → tensor conversion → channel-wise normalization.

- **Stage 1 — Uniform resizing:** every image is resized to 224×224 pixels by bilinear interpolation, removing native-resolution heterogeneity and ensuring compatibility with ImageNet-pretrained architectures.
- **Stage 2 — Split-conditional augmentation:** training images proceed through the stochastic augmentation sequence in Step 5; validation and test images bypass augmentation entirely.
- **Stage 3 — Tensor conversion:** transforms.ToTensor() reorders dimensions to the PyTorch convention and rescales pixel values from [0,255] to [0.0,1.0]. Grayscale images are broadcast to three channels.

- **Stage 4 — Channel-wise normalization:** `transforms.Normalize(mean=[0.5,0.5,0.5], std=[0.5,0.5,0.5])` maps the $[0.0,1.0]$ range to $[-1.0,1.0]$. Identical parameters are applied to all three branches.

5. Data augmentation (training split only)

Five stochastic operations, chosen to simulate realistic MRI acquisition variability while preserving diagnostic signal, are applied exclusively to the training split:

- **Random rotation ($\pm 15^\circ$):** simulates minor head tilt or scanner misalignment.
- **Random horizontal flip ($p = 0.5$):** accounts for approximate left-right anatomical symmetry.
- **Random vertical flip ($p = 0.2$):** applied at reduced probability, since the brain lacks comparable superior-inferior symmetry.
- **Random affine translation ($\pm 10\%$):** simulates imperfect centering within the imaging field of view.
- **Randomly resized crop (scale 0.8–1.0):** simulates variable zoom while preserving the 224×224 output size.

Intensity-based perturbations (brightness jitter, noise injection, colour jitter) are deliberately excluded, so every augmented sample remains a physically plausible variation of the clinical imaging protocol. Figure 2 shows each operation applied individually to the same representative training image.

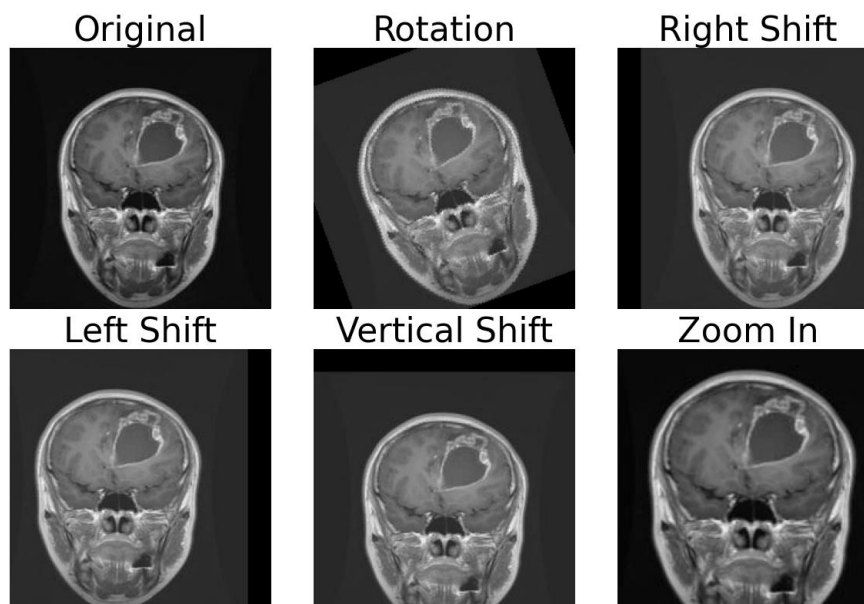


Figure 2. The five augmentation operations applied individually to the same coronal T1-weighted glioma training image (224×224 pixels after resizing).

6. Applying the method to a new clinical pool

To reproduce this method on a different clinical image pool: (a) collect and anonymize images under appropriate institutional ethics clearance, organizing them by diagnostic label; (b) compute an MD5 hash for every image and remove byte-identical duplicates; (c) compute a perceptual hash for every remaining image and remove pairs below a chosen Hamming-distance threshold (5 was used here); (d) perform per-class stratified splitting into training/validation/test sets; (e) run a content-hash comparison across all split pairs to confirm zero overlap before any model development begins; (f) apply the `resize` → `split-conditional-augment` → `tensor-convert` → `normalize`

pipeline, adjusting resolution, augmentation set, and normalization constants to the target modality and downstream architecture.

Method validation

The method was validated in two ways: (i) confirming that duplicate screening materially changes the native characteristics of the released pool, and (ii) confirming that the resulting duplicate-cleaned splits remain learnable by benchmarking seven deep-learning architectures under identical preprocessing and augmentation conditions.

Native image characterization

Tables 2 and 3 characterize the native properties of the 5,941 images retained after Step 2, computed across the full released set (not a sample), giving downstream users a precise description of what they will encounter before any preprocessing is applied.

Class	Width mean	Height mean	Size mean (KB)	Min (KB)	Median (KB)	Max (KB)
Glioma	405.50	406.02	15.71	3.31	16.32	70.21
Meningioma	416.82	418.31	18.34	4.08	19.21	227.59
No Tumor	290.62	293.67	19.50	3.39	11.93	275.97
Pituitary	416.33	416.31	23.54	4.87	25.43	109.96

Table 2. Per-class native width/height/file-size statistics across the full 5,941-image dataset, computed before preprocessing.

Width (px)	Height (px)	Count (%)
512	512	3297 (55.5%)
224	224	1958 (33.0%)
225	225	73 (1.2%)
236	236	27 (0.5%)
256	256	22 (0.4%)

Table 3. Five most frequent native image resolutions across the full dataset. 512×512 and 224×224 together account for 88.5% of all images, confirming that the uniform-resizing stage of the preprocessing pipeline (Method details, Step 4) is necessary before model use.

Figure 3 shows the per-class and per-split counts produced by Step 3, together with overall class distribution, per-class split proportions, and a class×split heatmap, confirming stratified allocation was achieved in practice.

Brain Tumor MRI Dataset — Final 70/15/15 Split

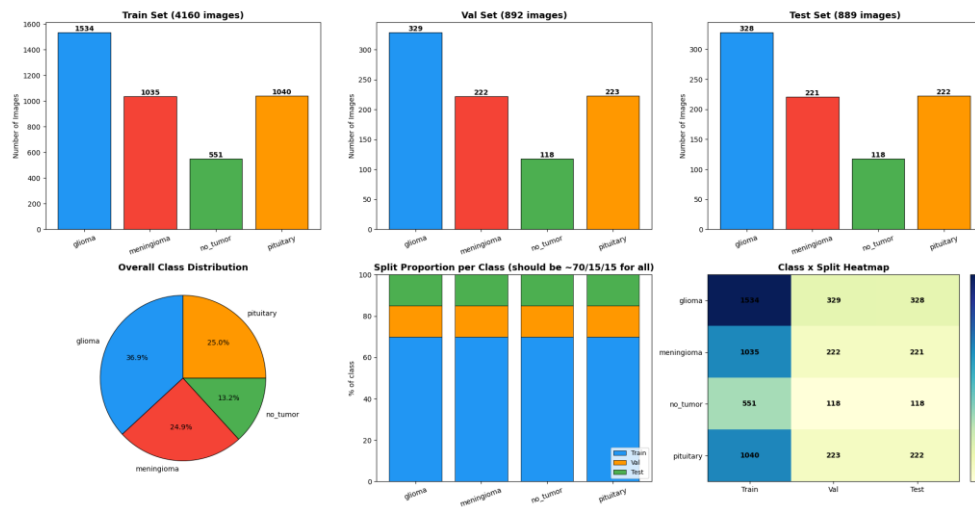


Figure 3. Class distribution analysis of the final duplicate-cleaned, 70/15/15-split dataset (5,941 images). Relative class frequencies are consistent across all three splits.

Figure 4 shows one representative image per class from the curated pool, and Figures 5–6 report pixel-intensity and quality-metric summaries computed on the curated, duplicate-cleaned images, confirming that the screening step did not introduce systematic quality artefacts.

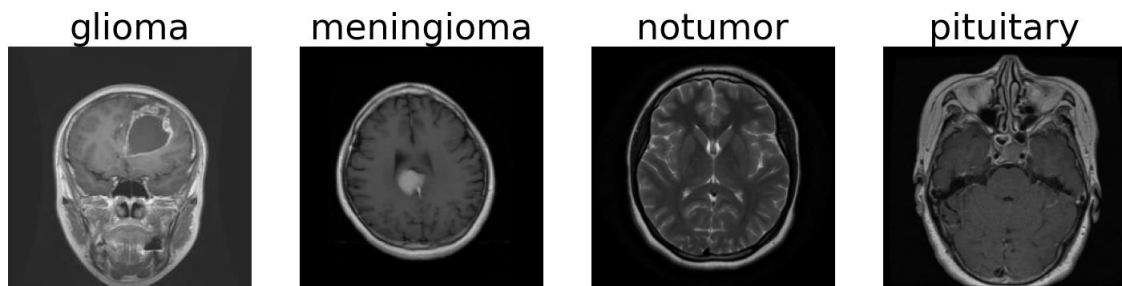


Figure 4. Representative raw MRI samples from each of the four diagnostic classes, shown at native resolution and contrast.

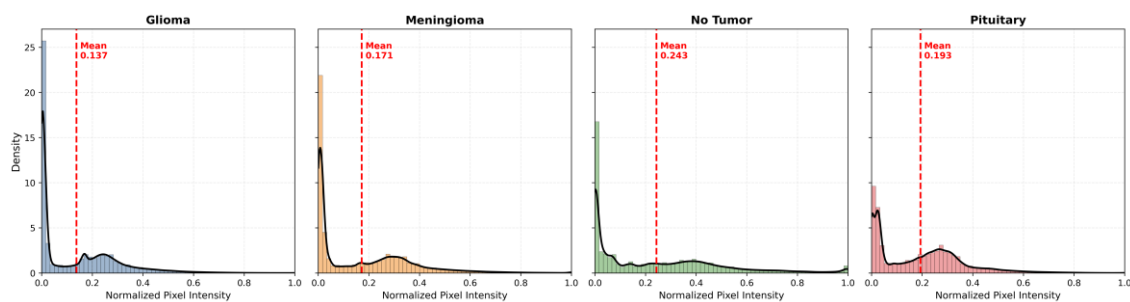


Figure 5. Pixel-intensity histograms (normalized to [0,1]) computed per class from 100 grayscale training images, resized to 224×224. All classes share a low-intensity spike near zero (background).

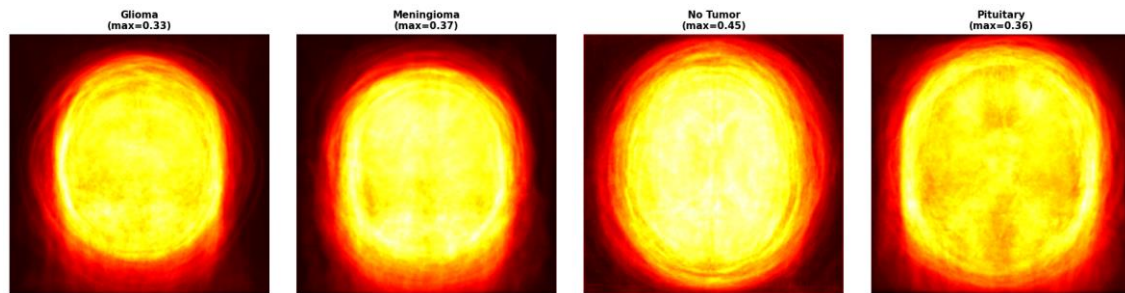


Figure 6. Class-wise mean MRI images computed by pixel-wise averaging of 100 grayscale training samples per class, displayed with a “hot” colormap.

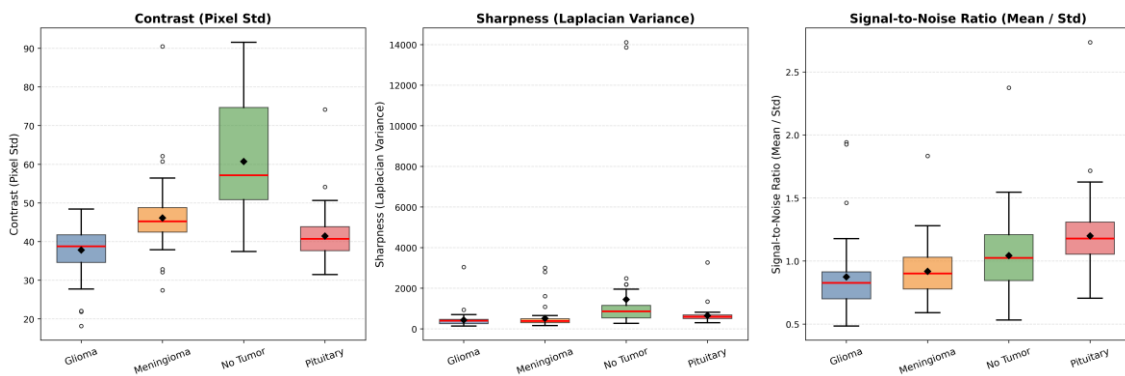


Figure 7. Boxplots of contrast, sharpness, and SNR per class from 50 randomly sampled training images. No class shows pathological quality degradation.

Downstream benchmarking

To validate that the curated, duplicate-cleaned splits remain learnable and are free of leakage-inflated performance, seven representative deep-learning architectures were trained and evaluated under identical preprocessing and augmentation conditions: a custom CNN, a Vision Transformer (ViT), and a Hybrid CNN–ViT trained from scratch; and ResNet50 and DenseNet121 under both frozen and fine-tuned pretrained configurations.

These seven architectures are deliberately trained with standard, unmodified deep-learning pipelines — conventional supervised training, off-the-shelf ImageNet-pretrained backbones, and no architecture-specific tuning beyond what is documented in Method details — because their role here is diagnostic rather than novel. A conventional, unmodified pipeline is used as a fixed external instrument to test whether the duplicate-screened, leakage-verified splits are learnable and free of leakage-inflated performance; it is deliberately not presented as a methodological contribution in itself, and no claim of architectural novelty is made for this stage. The methodological novelty of this paper lies entirely upstream of benchmarking, in the duplicate/near-duplicate screening and post-split leakage-verification protocol (Method details, Steps 2–3). Keeping the downstream benchmarking stage conventional is intentional: it ensures that the reported accuracy gains reflect the quality of the curated data rather than any architectural innovation, which would otherwise confound the validation.

Architecture	Paradigm	Test Acc.	Test F1
CNN	From scratch	0.8470	0.8457

ViT	From scratch	0.8256	0.8240
Hybrid CNN–ViT	From scratch	0.8189	0.8201
ResNet50	Frozen pretrained	0.8031	0.8022
DenseNet121	Frozen pretrained	0.8706	0.8703
ResNet50	Fine-tuned	0.9708	0.9708
DenseNet121	Fine-tuned	0.9831 (best)	0.9831 (best)

Table 4. Baseline classification performance of seven deep-learning architectures on the held-out test set (889 images). Fine-tuned DenseNet121 achieved the best result (98.31% accuracy).

Fine-tuned pretrained models achieved the strongest performance (DenseNet121: 98.31% accuracy; ResNet50: 97.08%), and frozen DenseNet121 (87.06%) outperformed the from-scratch CNN (84.70%), while frozen ResNet50 (80.31%) trailed both. This ordering fine-tuned > frozen pretrained $\approx$ from-scratch CNN > from-scratch transformer is consistent with expectations for a leakage-free dataset of this scale, supporting the conclusion that the duplicate-screening and splitting method (Steps 2–3) removed hidden leakage without leaving the resulting task trivially easy or unlearnable.

Limitations

The method was demonstrated on images from a single clinical facility (Epic Health Care, Chittagong, Bangladesh) under a fixed set of scanner protocols; datasets curated with this method from a single site may still require recalibration for models to generalize to other centres with different field strengths, scanner manufacturers, or contrast protocols. The perceptual-hash near-duplicate threshold (Hamming distance ≤ 5) was chosen empirically for this pool and may need adjustment for pools with different native resolutions or compression characteristics. The stratified splitting step preserves per-class proportions but does not correct pre-existing class imbalance in the source pool (in this demonstration, glioma: 36.9% vs. no tumor: 13.2%, ratio $\approx 2.78:1$); users applying the method to imbalanced pools should still consider class-weighted loss functions or stratified batch sampling downstream. The method as demonstrated screens and labels images at the whole-image level only; it does not include a step for generating pixel-level segmentation masks or bounding-box annotations, and MRI sequence metadata (T1, T2, post-contrast) is not captured by the pipeline unless separately recorded at the sourcing stage.

Ethics statements

This work involved human subjects. All MRI images were prospectively collected from Epic Health Care, Chittagong, Bangladesh, under institutional ethics clearance (Ref: EHC/CARD/2025/0047), conducted in accordance with the ethical principles of the Declaration of Helsinki. Informed consent was obtained from all patients prior to data collection. All images were fully anonymized prior to handover, with no personally identifiable information retained, in strict compliance with institutional data-protection and patient-confidentiality obligations. The anonymized dataset produced by this method is released for open-access academic use under the terms approved by the ethics committee.

CRediT author statement

Md Irfanul Kabir Hira: Conceptualization, Data curation, Methodology, Software, Formal analysis, Visualization, Writing – original draft. **Umme Sara:** Data curation, Validation, Writing – review & editing. **Mst Moriom Akter Bithee:** Data curation, Validation, Writing – review & editing. **Md Sohag Hossain:** Data curation, Software, Formal analysis, Writing – review & editing. **Md. Kowsar Ahmed:** Data curation, Validation, Writing – review & editing.



Abdullah Al Towsif: Data curation, Software, Writing – review & editing. **Md Mahmudul Hasan:** Data curation, Validation, Writing – review & editing. **Mohammad Shorif Uddin:** Supervision, Project administration, Writing – review & editing.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

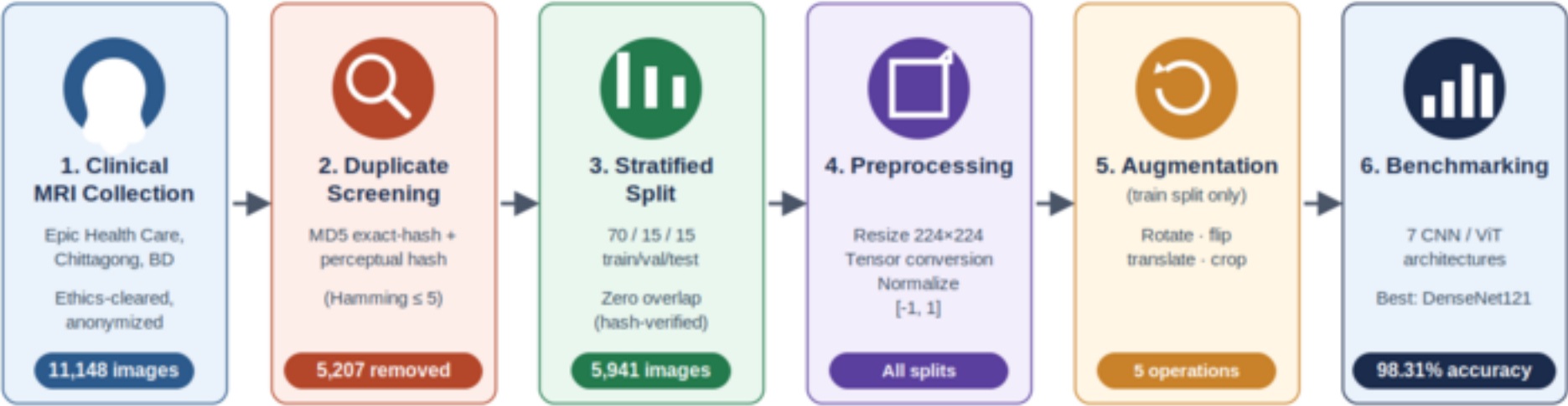
References

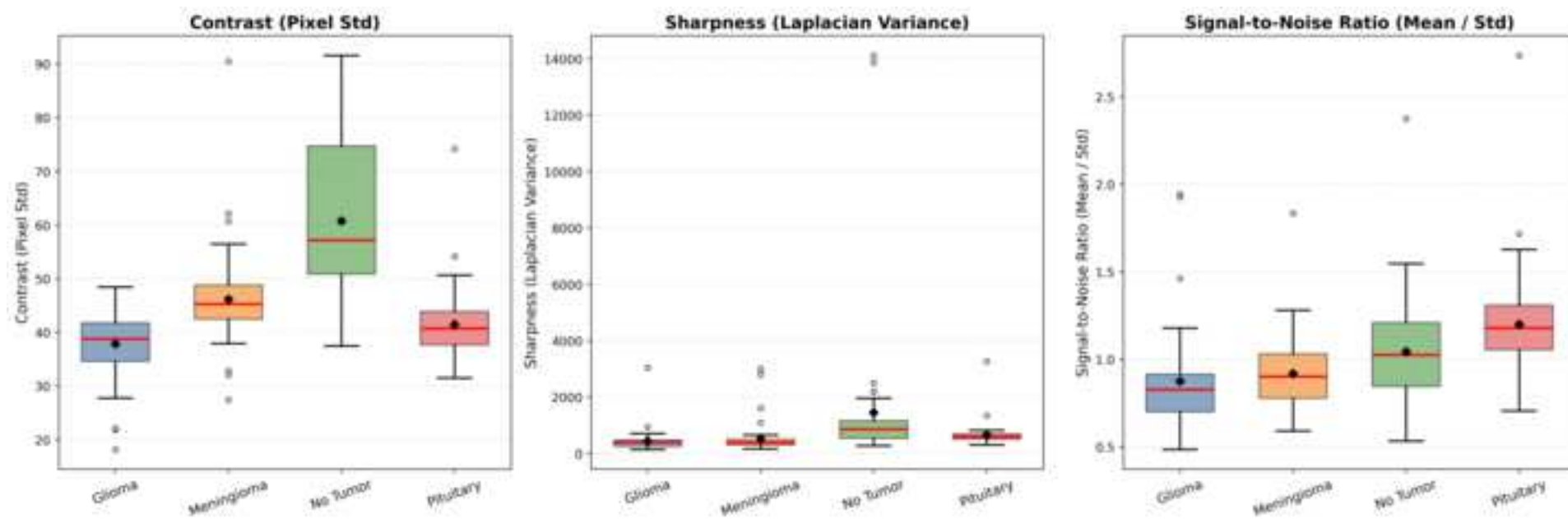
- [1] R. Disci, F. Gurcan, A. Soylu, Advanced brain tumor classification in MR images using transfer learning and pre-trained deep CNN models, *Cancers* 17 (2025) 121. doi:[10.3390/cancers17010121](https://doi.org/10.3390/cancers17010121).
- [2] R. Nimmagadda, P.K. Devi, A deep learning approach for brain tumour classification and detection in MRI images using YOLOv7, *Frontiers in Oncology* 15 (2025) 1508326. doi:[10.3389/fonc.2025.1508326](https://doi.org/10.3389/fonc.2025.1508326).
- [3] E. Zarenia, A.A. Far, K. Rezaee, Automated multi-class MRI brain tumor classification and segmentation using deformable attention and saliency mapping, *Scientific Reports* 15 (2025) 8114. doi:[10.1038/s41598-025-92776-1](https://doi.org/10.1038/s41598-025-92776-1).
- [4] V. Satushe, V. Vyas, S. Metkar, D.P. Singh, AI in MRI brain tumor diagnosis: A systematic review of machine learning and deep learning advances (2010–2025), *Chemometrics and Intelligent Laboratory Systems* 263 (2025) 105414. doi:[10.1016/j.chemolab.2025.105414](https://doi.org/10.1016/j.chemolab.2025.105414).
- [5] G. Al-Rumaihi, M.M. Khan, A. Saleh, A. Ali, L. Al-Romaihi, N. Al-Jaber, G. Al-Sulaiti, M.E.H. Chowdhury, G.R. Babu, S. Pedersen, Performance evaluation of artificial intelligence techniques in the diagnosis of brain tumors: A systematic review and meta-analysis, *Cureus* (2025). doi:[10.7759/cureus.88915](https://doi.org/10.7759/cureus.88915).
- [6] N. Netshamutshedzi, R. Netshikweta, J.-C. Ndogmo, I.C. Obagbuwa, A systematic review of the hybrid machine learning models for brain tumour segmentation and detection in medical images, *Frontiers in Artificial Intelligence* 8 (2025) 1615550. doi:[10.3389/frai.2025.1615550](https://doi.org/10.3389/frai.2025.1615550).
- [7] A. Fateh, Y. Rezvani, S. Moayedi, S. Rezvani, F. Fateh, M. Fateh, V. Abolghasemi, BRISC: Annotated dataset for brain tumor segmentation and classification, *Scientific Data* (2026). doi:[10.1038/s41597-026-06753-y](https://doi.org/10.1038/s41597-026-06753-y).
- [8] A.J. Aiya, N. Wani, Optimized deep learning for brain tumor detection: a hybrid approach with attention mechanisms and clinical explainability, *Scientific Reports* 15 (2025) 31386. doi:[10.1038/s41598-025-04591-3](https://doi.org/10.1038/s41598-025-04591-3).

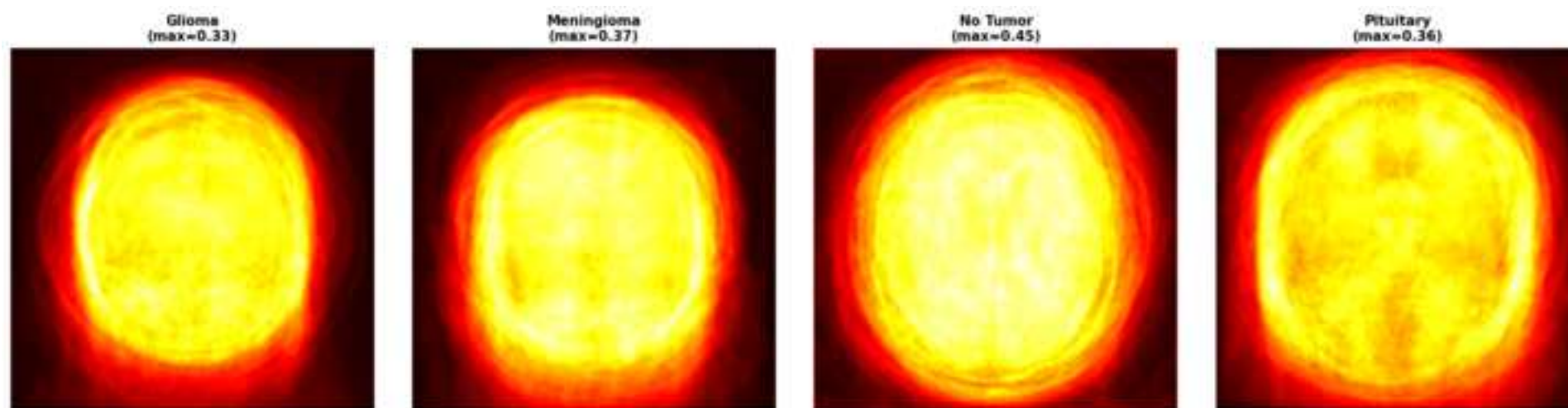


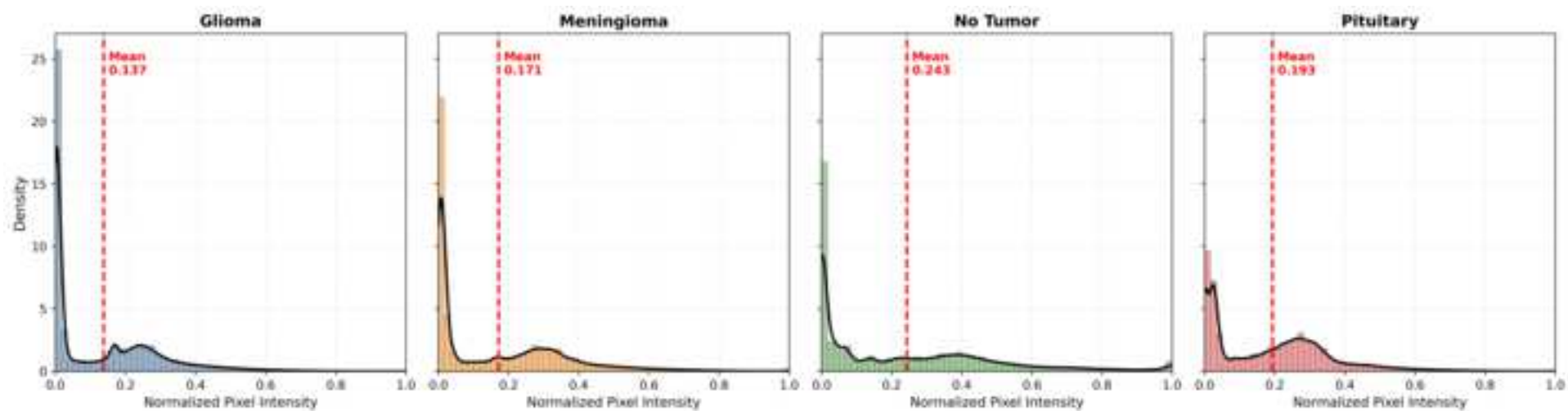
- [9] G. Jayaraman, S. Meganathan, S.S.M. Shah, M. Anuradha, R.K. Sundararajan, R. Rajakumar, A hybrid CNN–ViT framework with cross-attention fusion and data augmentation for robust brain tumor classification, Scientific Reports 15 (2025) 45769. doi:[10.1038/s41598-025-28636-9](https://doi.org/10.1038/s41598-025-28636-9).
- [10] Y. Wong, E.L.M. Su, C.F. Yeong, W. Holderbaum, C. Yang, Brain tumor classification using MRI images and deep learning techniques, PLOS ONE 20 (2025) e0322624. doi:[10.1371/journal.pone.0322624](https://doi.org/10.1371/journal.pone.0322624).
- [11] V.R. Srinivas, R. Parvathi, Explainable AI-driven MRI-based brain tumor classification: a novel deep learning approach, Frontiers in Artificial Intelligence 8 (2026) 1700214. doi:[10.3389/frai.2025.1700214](https://doi.org/10.3389/frai.2025.1700214).

BDNeuro-MRI: Duplicate-Screened Clinical MRI Curation Pipeline

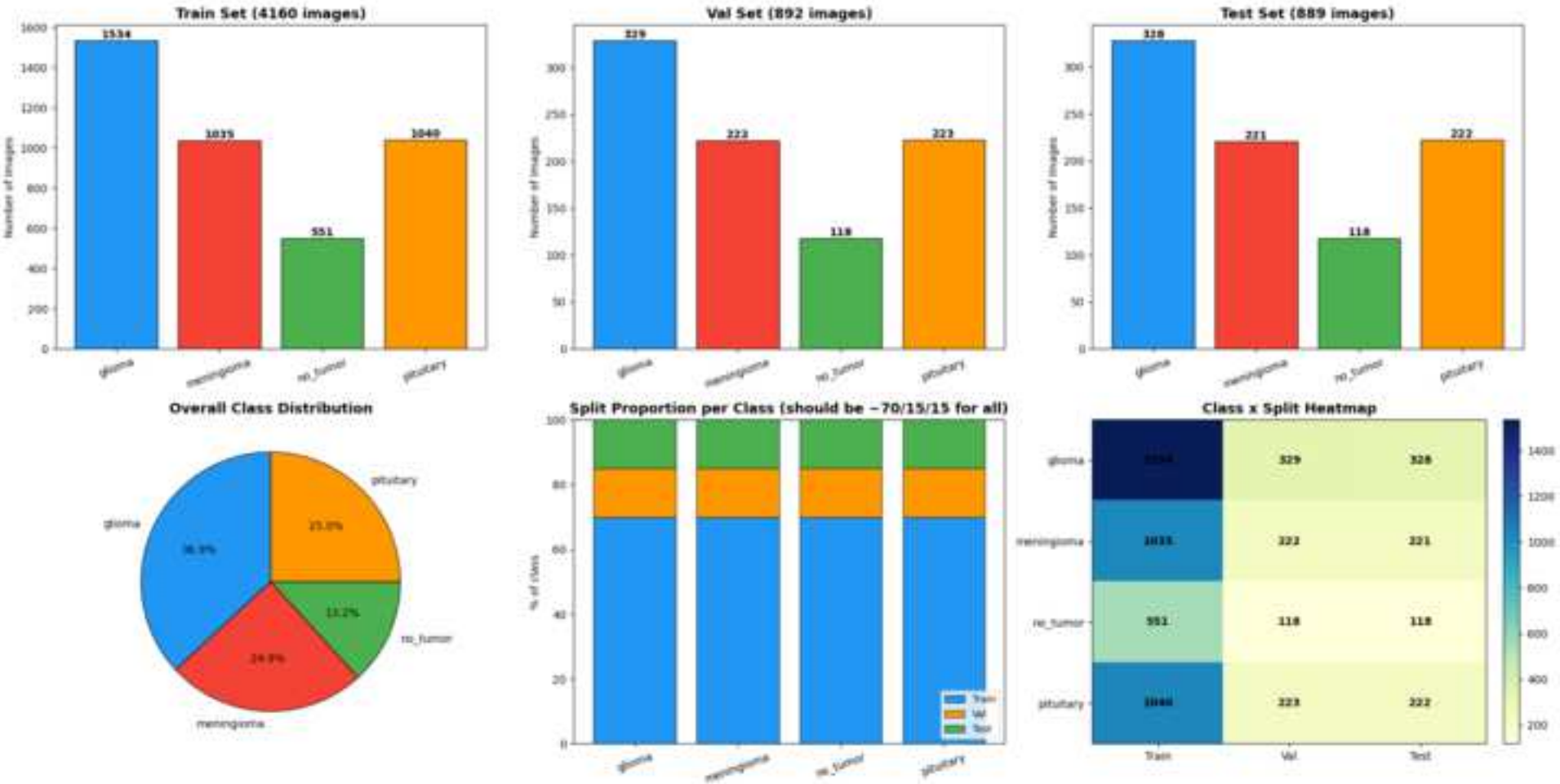




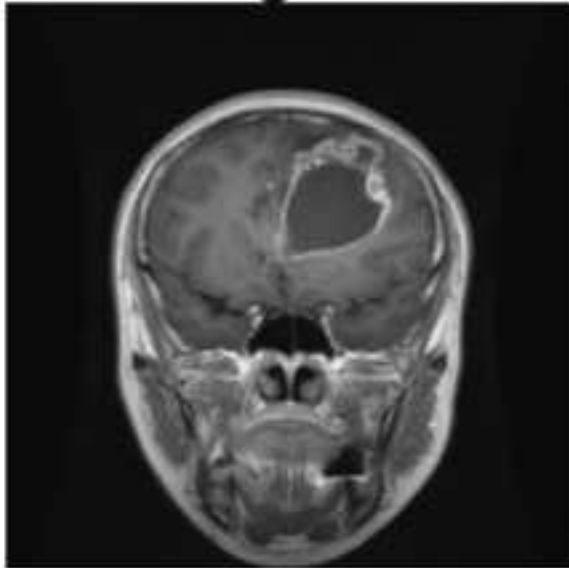




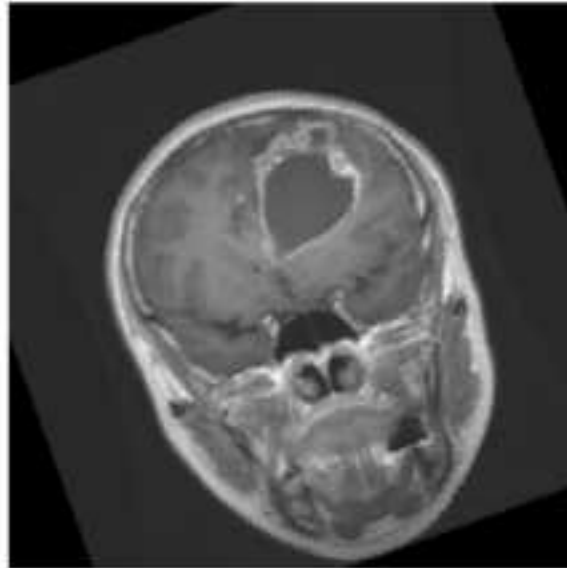
Brain Tumor MRI Dataset — Final 70/15/15 Split



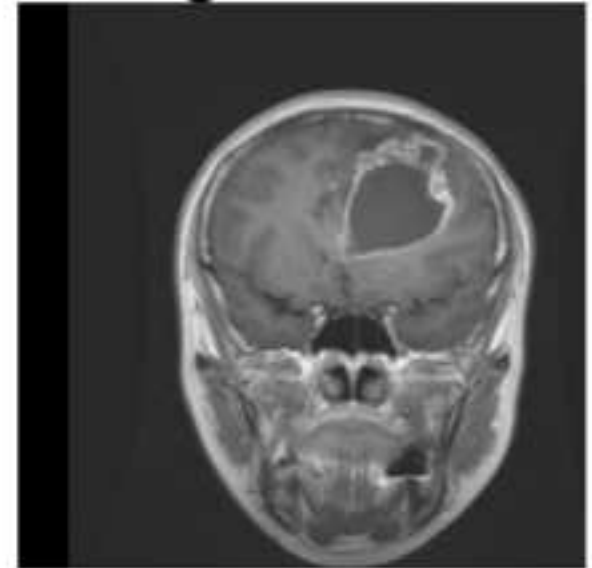
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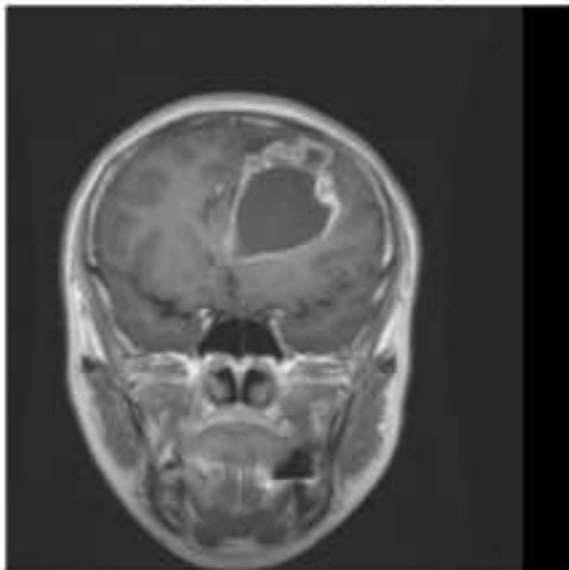
Rotation



Right Shift



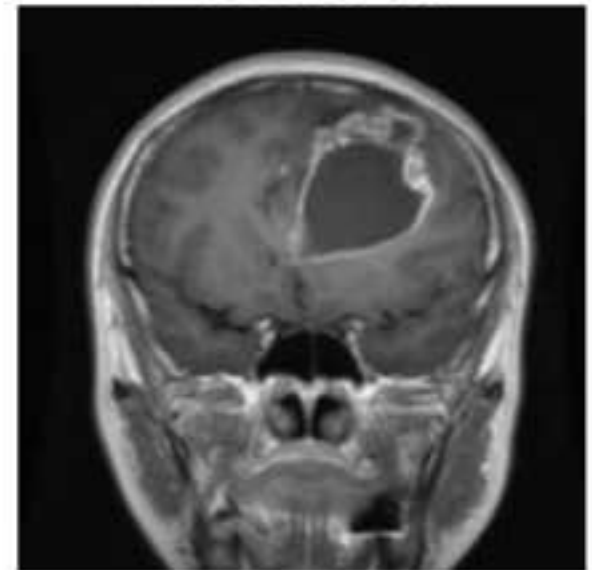
Left Shift

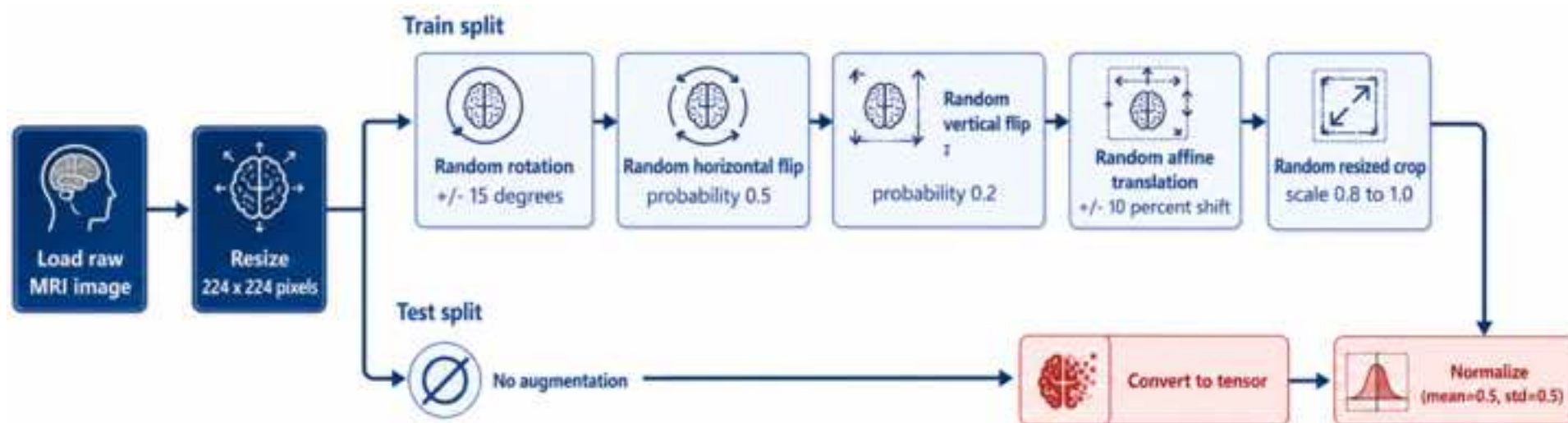


Vertical Shift

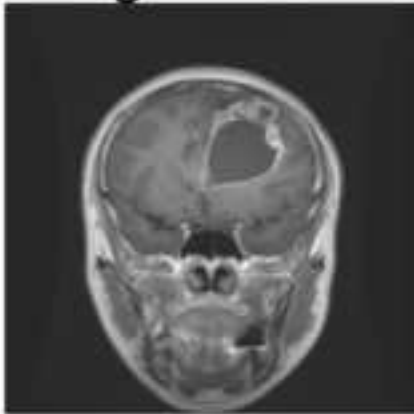


Zoom In

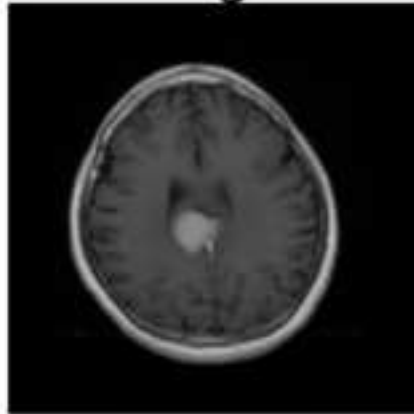




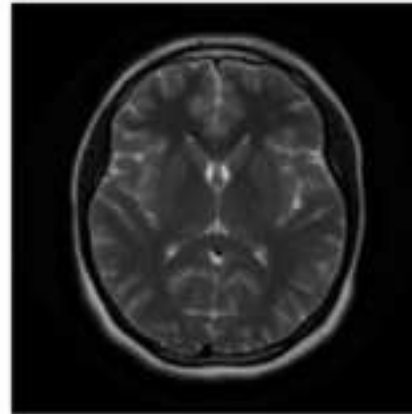
glioma



meningioma



notumor



pituitary

